

RAHUL KP

Software Engineer Intern

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EDUCATION

University of Calicut

MSc Computer Science | 2024–2026 · **CGPA: 77%**

BSc Computer Science | 2020–2023

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL

Core CS: Data Structures, Algorithms, OOP, REST APIs, Authentication, Authorization, Git, CI/CD

Backend & Web: FastAPI, Flask, Node.js, Express.js, React, Next.js

Databases & Infrastructure: PostgreSQL, Redis, Docker, Kubernetes, GitHub Actions

Observability & Deployment: Prometheus, Grafana, Vercel, Render

Machine Learning: PyTorch, NumPy, scikit-learn, Neural Networks, CNNs

PROJECTS

RankScript — Full-Stack Learning & Ranking Platform

FastAPI · **Next.js** · **PostgreSQL** · **Redis** · **Docker**

[GitHub](#) · [Live Demo](#)

- Engineered a full-stack platform with layered backend architecture separating API routes, business logic, and database models using **FastAPI, PostgreSQL, Redis, and Next.js**.
- Implemented **role-based access control** for student, mentor, and admin workflows with JWT authentication and refresh-token rotation.
- Built a weighted ranking engine combining quiz scores, assignments, completion rates, and activity streaks into **district, state, and national leaderboards**, using PostgreSQL materialized views for efficient pagination.
- Developed **266 automated backend tests** and containerized the application with Docker for reproducible deployment.

PhishGuard — Production ML API & MLOps System

Python · **Flask** · **scikit-learn** · **Docker** · **Kubernetes** · **Prometheus**

[GitHub](#) · [Live Demo](#)

- Engineered a production REST API with prediction and batch endpoints, input validation, request tracing, structured JSON logging, and Prometheus metrics for phishing-URL detection (~95% accuracy, 0.98 ROC-AUC).
- Implemented automated data-drift monitoring and Kubernetes deployment with **HPA and Kustomize** for horizontal autoscaling.
- Built GitHub Actions CI/CD with **CodeQL static analysis**, backed by **222 automated tests achieving 98.7% code coverage**.

TaskFlow — Full-Stack Kanban Task Management App

React · **Express** · **MongoDB**

[GitHub](#) · [Live Demo](#)

- Built and deployed a full-stack Kanban application with React/Vite, Express, and MongoDB, including drag-and-drop task management and role-based workflows.
- Diagnosed and resolved a cross-site cookie authentication failure caused by **Safari Intelligent Tracking Prevention**, re-architecting authentication through a same-origin Vercel API proxy.
- Resolved production **CORS** and deployment issues, including a silently failing database seeder on Render's native runtime.

CERTIFICATIONS

- IBM Generative AI Engineering Professional Certificate
- Duke University MLOps | Machine Learning Operations Specialization
- Google Cloud AI Infrastructure Specialization